

The Gödel Sentence for Peano Arithmetic

Anton Ivanov

<https://github.com/forall-x/godel-sentence>

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1 Preliminaries: Arithmetic Dialect

The following construction is formulated in First-Order Peano Arithmetic with the signature $\{0, S, +, \cdot, =\}$.

1.1 Document Notation: Decimal Shorthand

For legibility, formal successor chains are rendered in this document as decimal integers (e.g., $S(S(0)) \equiv 2$). This is a presentation shorthand and does not alter the underlying formal terms.

1.2 Algebraic Simplifications

The following logical identities are applied during AST construction:

- **Additive Identity:** $0 + t \equiv t$ and $t + 0 \equiv t$
- **Multiplicative Identity:** $1 \cdot t \equiv t$ and $t \cdot 1 \equiv t$
- **Zero Law:** $0 \cdot t \equiv 0$ and $t \cdot 0 \equiv 0$

1.3 Formal Arithmetization: Positional Numerals

The formal arithmetization requires representing Gödel numbers of $n \approx 10^{7296}$, for which unary chains are infeasible. We use the positional encoding formula $\text{Num}(v, n)$, provably equivalent to $v = n$, defining v via the bit-doubling identity $v_{i+1} = (v_i + v_i) + b_i$. This reduces formula character complexity from $O(n)$ to $O(\log n)$.

For example, $\text{Num}(v, 5)$ (binary 101_2) is constructed by binding intermediate doubling steps with existential quantifiers:

$$\exists v_2 (\exists v_1 (\exists v_0 (v_0 = S(0) \wedge v_1 = v_0 + v_0) \wedge v_2 = S(v_1 + v_1)) \wedge v = v_2)$$

2 Formal Derivation

The Gödel sentence G is constructed via the Diagonal Lemma using a fixed-width block DAG encoding.

2.1 Unprovability Predicate

Let $\Psi(y)$ be the formula asserting that there exists no proof p of the formula with Gödel number y :

$$\Psi(y) \iff \neg \exists p \text{ IsProof}(p, y)$$

2.2 Diagonalization Template

Let $\alpha(x)$ be the formula asserting $\Psi(\text{diag}(x))$:

$$\alpha(x) \iff \exists x_{\text{diag}} (\text{diag}(x, x_{\text{diag}}) \wedge \Psi(x_{\text{diag}}))$$

2.3 Fixed Point Calculation

Let n be the Gödel number of $\alpha(x)$:

$$n = gn(\alpha) = 69527427086362971181 \dots (24240 \text{ bits})$$

2.4 The Gödel Sentence

The sentence G is the diagonalization of α , specifying the value $n = gn(\alpha)$ via the Num predicate:

$$G \iff \exists n_x (\text{Num}(n_x, n) \wedge \alpha(n_x))$$

3 The Complete Gödel Sentence

The full expanded form of G is provided below.

$G = \exists v_0 (\exists v_2 \exists v_3 \exists v_4 ((\exists v_5 ((28734291391235416102526681732724021398500152667694694$
 $6 + S(v_5)) = S(v_3)) \wedge \exists v_6 (v_2 = ((v_6 \cdot S(v_3)) + 28734291391235416102526681732724021398500152667694694$
 $6))) \wedge ((\exists v_7 ((v_0 + S(v_7)) = S((S(v_4) \cdot v_3))) \wedge \exists v_8 (v_2 = ((v_8 \cdot S((S(v_4) \cdot v_3))) + v_0))) \wedge \forall v_9 ((\exists v_{10} ((v_9 + S(v_{10})) = v_4) \vee (v_9 =$
 $v_4)) \rightarrow (\exists v_{11} ((v_9 + S(v_{11})) = v_4) \rightarrow \exists v_{12} \exists v_{13} \exists v_{14} ((\exists v_{15} ((v_{12} + S(v_{15})) = S((S(v_9) \cdot v_3))) \wedge \exists v_{16} (v_2 =$
 $((v_{16} \cdot S((S(v_9) \cdot v_3))) + v_{12}))) \wedge ((\exists v_{17} ((v_{13} + S(v_{17})) = S((S(v_9) \cdot v_3))) \wedge \exists v_{18} (v_2 = ((v_{18} \cdot S((S(v_9) \cdot v_3))) + v_{13}))) \wedge$
 $(\exists v_{19} \exists v_{20} \exists v_{21} (\exists v_{22} \exists v_{23} ((\exists v_{24} ((1 + S(v_{24})) = S(v_{23})) \wedge \exists v_{25} (v_{22} = ((v_{25} \cdot S(v_{23})) + 1))) \wedge ((\exists v_{26} ((v_{19} + S(v_{26})) =$
 $S((S(v_9) \cdot v_{23}))) \wedge \exists v_{27} (v_{22} = ((v_{27} \cdot S((S(v_9) \cdot v_{23}))) + v_{19}))) \wedge \forall v_{28} (\exists v_{29} ((v_{28} + S(v_{29})) = v_9) \rightarrow$
 $\exists v_{30} \exists v_{31} ((\exists v_{32} ((v_{30} + S(v_{32})) = S((S(v_{28}) \cdot v_{23}))) \wedge \exists v_{33} (v_{22} = ((v_{33} \cdot S((S(v_{28}) \cdot v_{23}))) + v_{30}))) \wedge ((\exists v_{34} ((v_{31} + S(v_{34})) =$
 $S((S(S(v_{28})) \cdot v_{23}))) \wedge \exists v_{35} (v_{22} = ((v_{35} \cdot S((S(S(v_{28})) \cdot v_{23}))) + v_{31}))) \wedge (v_{31} = (v_{30} + v_{30})))))) \wedge ((69527427086362971181030$
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66768351199635 = ((v₂₀ · (v₁₉ · 2)) + ((v₁₄ · v₁₉ + v₂₁))) ∧ ∃v₃₆((v₂₁ + S(v₃₆)) = v₁₉)) ∧ (((v₁₄ = 0) → v₁₃ =
(v₁₂ + v₁₂))) ∧ (¬(v₁₄ = 0) → v₁₃ = S((v₁₂ + v₁₂))))))))) ∧ ∃v₃₇((∃v₃₈∃v₃₉(∃v₄₀∃v₄₁∃v₄₂((∃v₄₃((28734291391235416102526
6817327240213985001526676946946 + S(v₄₃)) = S(v₄₁)) ∧ ∃v₄₄(v₄₀ = ((v₄₄ · S(v₄₁)) + 2873429139123541610252668173272402
1398500152667694694
6))) ∧ ((∃v₄₅((v₃₉ + S(v₄₅)) = S((S(v₄₂) · v₄₁))) ∧ ∃v₄₆(v₄₀ = ((v₄₆ · S((S(v₄₂) · v₄₁))) + v₃₉))) ∧ ∀v₄₇((∃v₄₈((v₄₇ + S(v₄₈)) =
v₄₂) ∨ (v₄₇ = v₄₂)) → (∃v₄₉((v₄₇ + S(v₄₉)) = v₄₂) → ∃v₅₀∃v₅₁∃v₅₂((∃v₅₃((v₅₀ + S(v₅₃)) = S((S(v₄₇) · v₄₁))) ∧ ∃v₅₄(v₄₀ =
((v₅₄ · S((S(v₄₇) · v₄₁))) + v₅₀))) ∧ ((∃v₅₅((v₅₁ + S(v₅₅)) = S((S(v₄₇) · v₄₁))) ∧ ∃v₅₆(v₄₀ = ((v₅₆ · S((S(v₄₇) · v₄₁))) +
v₅₁))) ∧ (∃v₅₇∃v₅₈∃v₅₉(∃v₆₀∃v₆₁((∃v₆₂((1 + S(v₆₂)) = S(v₆₁)) ∧ ∃v₆₃(v₆₀ = ((v₆₃ · S(v₆₁)) + 1))) ∧ ((∃v₆₄((v₅₇ + S(v₆₄)) =
S((S(v₄₇) · v₆₁))) ∧ ∃v₆₅(v₆₀ = ((v₆₅ · S((S(v₄₇) · v₆₁))) + v₅₇))) ∧ ∀v₆₆((∃v₆₇((v₆₆ + S(v₆₇)) = v₄₇) →
∃v₆₈∃v₆₉((∃v₇₀((v₆₈ + S(v₇₀)) = S((S(v₆₆) · v₆₁))) ∧ ∃v₇₁(v₆₀ = ((v₇₁ · S((S(v₆₆) · v₆₁))) + v₆₈))) ∧ ((∃v₇₂((v₆₉ + S(v₇₂)) =
S((S(v₆₆) · v₆₁))) ∧ ∃v₇₃(v₆₀ = ((v₇₃ · S((S(v₆₆) · v₆₁))) + v₆₉))) ∧ (v₆₉ = (v₆₈ + v₆₈))))))))) ∧ ((v₀ =
((v₅₈ · (v₅₇ · 2)) + ((v₅₂ · v₅₇ + v₅₉))) ∧ ∃v₇₄((v₅₉ + S(v₇₄)) = v₅₇))) ∧ (((v₅₂ = 0) → v₅₁ = (v₅₀ + v₅₀))) ∧ (¬(v₅₂ = 0) →
(v₅₁ = S((v₅₀ + v₅₀))))))))) ∧ ∃v₇₅∃v₇₆∃v₇₇(∃v₇₈∃v₇₉∃v₈₀(∃v₈₁∃v₈₂((∃v₈₃((1 + S(v₈₃)) = S(v₈₂)) ∧ ∃v₈₄(v₈₁ =
((v₈₄ · S(v₈₂)) + 1))) ∧ ((∃v₈₅((v₇₈ + S(v₈₅)) = S(v₈₂)) ∧ ∃v₈₆(v₈₁ = ((v₈₆ · S(v₈₂)) + v₇₈))) ∧ ∀v₈₇(∃v₈₈((v₈₇ + S(v₈₈)) = 0) →
∃v₈₉∃v₉₀((∃v₉₁((v₈₉ + S(v₉₁)) = S((S(v₈₇) · v₈₂))) ∧ ∃v₉₂(v₈₁ = ((v₉₂ · S((S(v₈₇) · v₈₂))) + v₈₉))) ∧ ((∃v₉₃((v₉₀ + S(v₉₃)) =
S((S(v₈₇) · v₈₂))) ∧ ∃v₉₄(v₈₁ = ((v₉₄ · S((S(v₈₇) · v₈₂))) + v₉₀))) ∧ (v₉₀ = (v₈₉ + v₈₉))))))))) ∧ ((v₀ = ((v₇₉ · v₇₈ · 11529215
04606846976)) + ((v₇₅ · v₇₈ + v₈₀))) ∧ ∃v₉₅((v₈₀ + S(v₉₅)) = v₇₈))) ∧ (∃v₉₆∃v₉₇∃v₉₈(∃v₉₉∃v₁₀₀((∃v₁₀₁((1 + S(v₁₀₁)) =
S(v₁₀₀)) ∧ ∃v₁₀₂(v₉₉ = (v₁₀₂ · S(v₁₀₀)) + 1))) ∧ ((∃v₁₀₃(v₉₆ + S(v₁₀₃)) = S(v₁₀₀)) ∧ ∃v₁₀₄(v₉₉ =
((v₁₀₄ · S(v₁₀₀)) + v₉₆))) ∧ ∀v₁₀₅(∃v₁₀₆((v₁₀₅ + S(v₁₀₆)) = 0) → ∃v₁₀₇∃v₁₀₈((∃v₁₀₉((v₁₀₇ + S(v₁₀₉)) =
S((S(v₁₀₅) · v₁₀₀)) ∧ ∃v₁₁₀(v₉₉ = ((v₁₁₀ · S((S(v₁₀₅) · v₁₀₀))) + v₁₀₇))) ∧ ((∃v₁₁₁((v₁₀₈ + S(v₁₁₁)) =
S((S(v₁₀₅) · v₁₀₀)) ∧ ∃v₁₁₂(v₉₉ = ((v₁₁₂ · S((S(v₁₀₅) · v₁₀₀))) + v₁₀₈))) ∧ (v₁₀₈ = (v₁₀₇ + v₁₀₇))))))))) ∧ ((v₃₈ = ((v₉₇ · v₉₆ · 1
152921504606846976)) + ((v₇₅ · v₉₆ + v₉₈))) ∧ ∃v₁₁₃((v₉₈ + S(v₁₁₃)) = v₉₆))) ∧ (∃v₁₁₄∃v₁₁₅((∃v₁₁₆((1 + S(v₁₁₆)) =
S(v₁₁₅)) ∧ ∃v₁₁₇(v₁₁₄ = ((v₁₁₇ · S(v₁₁₅)) + 1))) ∧ ((∃v₁₁₈((v₇₆ + S(v₁₁₈)) = S((29 · v₁₁₅))) ∧ ∃v₁₁₉(v₁₁₄ = ((v₁₁₉ · S((2
9 · v₁₁₅))) + v₇₆))) ∧ ∀v₁₂₀(∃v₁₂₁((v₁₂₀ + S(v₁₂₁)) = 2
8) → ∃v₁₂₂∃v₁₂₃((∃v₁₂₄((v₁₂₂ + S(v₁₂₄)) = S((S(v₁₂₀) · v₁₁₅))) ∧ ∃v₁₂₅(v₁₁₄ =
((v₁₂₅ · S((S(v₁₂₀) · v₁₁₅))) + v₁₂₂))) ∧ ((∃v₁₂₆((v₁₂₃ + S(v₁₂₆)) = S((S(v₁₂₀) · v₁₁₅))) ∧ ∃v₁₂₇(v₁₁₄ =
((v₁₂₇ · S((S(v₁₂₀) · v₁₁₅))) + v₁₂₃))) ∧ (v₁₂₃ = (v₁₂₂ + v₁₂₂))))))))) ∧ (∃v₁₂₈∃v₁₂₉((∃v₁₃₀((1 + S(v₁₃₀)) =
S(v₁₂₉)) ∧ ∃v₁₃₁(v₁₂₈ = ((v₁₃₁ · S(v₁₂₉)) + 1))) ∧ ((∃v₁₃₂((v₇₇ + S(v₁₃₂)) = S((57 · v₁₂₉))) ∧ ∃v₁₃₃(v₁₂₈ = ((v₁₃₃ · S((5
7 · v₁₂₉))) + v₇₇))) ∧ ∀v₁₃₄(∃v₁₃₅((v₁₃₄ + S(v₁₃₅)) = 5
6) → ∃v₁₃₆∃v₁₃₇((∃v₁₃₈((v₁₃₆ + S(v₁₃₈)) = S((S(v₁₃₄) · v₁₂₉))) ∧ ∃v₁₃₉(v₁₂₈ =
((v₁₃₉ · S((S(v₁₃₄) · v₁₂₉))) + v₁₃₆))) ∧ ((∃v₁₄₀((v₁₃₇ + S(v₁₄₀)) = S((S(v₁₃₄) · v₁₂₉))) ∧ ∃v₁₄₁(v₁₂₈ =
((v₁₄₁ · S((S(v₁₃₄) · v₁₂₉))) + v₁₃₇))) ∧ (v₁₃₇ = (v₁₃₆ + v₁₃₆))))))))) ∧ ∀v₁₄₂((∃v₁₄₃((v₁₄₂ + S(v₁₄₃)) = v₇₅) ∨ (v₁₄₂ = v₇₅)) →
(∃v₁₄₄(S(v₁₄₄) = v₁₄₂) ∧ ∃v₁₄₅((v₁₄₂ + S(v₁₄₅)) = S(v₇₅))) → ∃v₁₄₆∃v₁₄₇(∃v₁₄₈∃v₁₄₉∃v₁₅₀(∃v₁₅₁∃v₁₅₂((∃v₁₅₃((1 + S(v₁₅₃)) =
S(v₁₅₂)) ∧ ∃v₁₅₄(v₁₅₁ = ((v₁₅₄ · S(v₁₅₂)) + 1))) ∧ ((∃v₁₅₅((v₁₄₈ + S(v₁₅₅)) = S((S(v₁₄₂ · 6
0) · v₁₅₂))) ∧ ∃v₁₅₆(v₁₅₁ = ((v₁₅₆ · S((S(v₁₄₂ · 60) · v₁₅₂))) + v₁₄₈))) ∧ ∀v₁₅₇(∃v₁₅₈((v₁₅₇ + S(v₁₅₈)) = (v₁₄₂ · 6
0) → ∃v₁₅₉∃v₁₆₀(∃v₁₆₁((v₁₅₉ + S(v₁₆₁)) = S((S(v₁₅₇) · v₁₅₂))) ∧ ∃v₁₆₂(v₁₅₁ =
((v₁₆₂ · S((S(v₁₅₇) · v₁₅₂))) + v₁₅₉))) ∧ ((∃v₁₆₃((v₁₆₀ + S(v₁₆₃)) = S((S(v₁₅₇) · v₁₅₂))) ∧ ∃v₁₆₄(v₁₅₁ =
((v₁₆₄ · S((S(v₁₅₇) · v₁₅₂))) + v₁₆₀))) ∧ (v₁₆₀ = (v₁₅₉ + v₁₅₉))))))))) ∧ ((v₀ = ((v₁₄₉ · v₁₄₈ · 115292150460684697
6)) + ((v₁₄₆ · v₁₄₈ + v₁₅₀))) ∧ ∃v₁₆₅((v₁₅₀ + S(v₁₆₅)) = v₁₄₈))) ∧ (∃v₁₆₆∃v₁₆₇∃v₁₆₈(∃v₁₆₉∃v₁₇₀((∃v₁₇₁((1 + S(v₁₇₁)) =
S(v₁₇₀)) ∧ ∃v₁₇₂(v₁₆₉ = ((v₁₇₂ · S(v₁₇₀)) + 1))) ∧ ((∃v₁₇₃((v₁₆₆ + S(v₁₇₃)) = S((S(v₁₄₂ · 6
0) · v₁₇₀))) ∧ ∃v₁₇₄(v₁₆₉ = ((v₁₇₄ · S((S(v₁₄₂ · 60) · v₁₇₀))) + v₁₆₆))) ∧ ∀v₁₇₅(∃v₁₇₆((v₁₇₅ + S(v₁₇₆)) = (v₁₄₂ · 6

$0)) \rightarrow \exists v_{177} \exists v_{178} ((\exists v_{179} ((v_{177} + S(v_{179})) = S((S(v_{175}) \cdot v_{170}))) \wedge \exists v_{180} (v_{169} =$
 $((v_{180} \cdot S((S(v_{175}) \cdot v_{170}))) + v_{177})) \wedge ((\exists v_{181} ((v_{178} + S(v_{181})) = S((S(v_{175}) \cdot v_{170}))) \wedge \exists v_{182} (v_{169} =$
 $((v_{182} \cdot S((S(v_{175}) \cdot v_{170}))) + v_{178})) \wedge (v_{178} = (v_{177} + v_{177})))))) \wedge ((v_{38} = ((v_{167} \cdot (v_{166} \cdot 115292150460684697$
 $6)) + (v_{147} \cdot v_{166} + v_{168}))) \wedge \exists v_{183} ((v_{168} + S(v_{183})) = v_{166})) \wedge \exists v_{184} \exists v_{185} \exists v_{186} \exists v_{187} ((v_{146} =$
 $((v_{184} \cdot v_{77}) + ((v_{185} \cdot v_{76} + v_{187}))) \wedge ((v_{147} = ((v_{184} \cdot v_{77}) + ((v_{186} \cdot v_{76} + v_{187}))) \wedge (((v_{184} = 2) \wedge (v_{185} = 7102280006813659$
 $6)) \rightarrow (v_{186} = v_{39})) \wedge (\neg((v_{184} = 2) \wedge (v_{185} = 7102280006813659$
 $6)) \rightarrow (v_{186} = v_{185}))))))))) \wedge \neg \exists v_{188} \exists v_{189} \exists v_{190} \exists v_{191} (((v_{188} + v_{188}) =$
 $((v_{189} + v_{190}) \cdot S((v_{189} + v_{190}))) + (v_{189} + v_{189})) \wedge ((\exists v_{192} ((v_{37} + S(v_{192})) = S((S(v_{191}) \cdot v_{190}))) \wedge \exists v_{193} (v_{189} =$
 $((v_{193} \cdot S((S(v_{191}) \cdot v_{190}))) + v_{37}))) \wedge \forall v_{194} ((\exists v_{195} ((v_{194} + S(v_{195})) = v_{191}) \vee (v_{194} = v_{191})) \rightarrow (\exists v_{196} (S(v_{196}) = v_{194}) \rightarrow$
 $\exists v_{197} ((\exists v_{198} ((v_{197} + S(v_{198})) = S((S(v_{194}) \cdot v_{190}))) \wedge \exists v_{199} (v_{189} = ((v_{199} \cdot S((S(v_{194}) \cdot v_{190}))) + v_{197}))) \wedge (((v_{197} = 10274873$
 $8160399008155279519597442830332591347987614826164712945477299344645327814822745246399430633427933594$
 $1) \vee ((v_{197} = 19510928468548500563497340397041373327480480328643262288247244150451947352283787268824623658123693$
 $95658441565822140738927603897440046927140880391) \vee ((v_{197} = 7638884920740010778244731722580064036095728761364308$
 $41474875040787631647950732815008530436) \vee ((v_{197} = 1170655708549013273701036860197339416006416686919924193835937$
 $147469577528813791863002089900606666145544590341246147089640923115596174120609382407) \vee ((v_{197} = 763888492074001$
 $079701748178585039753791826470675781764541576681217976790954939403881938948) \vee ((v_{197} = 117065570854901327387026$
 $71882076430574848909171996781569024160321286724655534199694254139101480843104007228588531360083630364845088366$
 $88411230215) \vee ((v_{197} = 574685827846688814027160316283759829078372495355019266) \vee ((v_{197} = 1951092845400117260775852$
 $02604967860186557073976240101516955574088254710446297786364106992735964471043456386267728639087600048437270662$
 $10267824135)))))) \vee (\exists v_{200} \exists v_{201} \exists v_{202} \exists v_{203} \exists v_{204} \exists v_{205} \exists v_{206} \exists v_{207} (\exists v_{208} \exists v_{209} \exists v_{210} (\exists v_{211} \exists v_{212} ((\exists v_{213} ((1 + S(v_{213})) =$
 $S(v_{212})) \wedge \exists v_{214} (v_{211} = ((v_{214} \cdot S(v_{212})) + 1))) \wedge ((\exists v_{215} ((v_{208} + S(v_{215})) = S(v_{212})) \wedge \exists v_{216} (v_{211} =$
 $((v_{216} \cdot S(v_{212})) + v_{208}))) \wedge \forall v_{217} (\exists v_{218} ((v_{217} + S(v_{218})) = 0) \rightarrow \exists v_{219} \exists v_{220} ((\exists v_{221} ((v_{219} + S(v_{221})) =$
 $S((S(v_{217}) \cdot v_{212}))) \wedge \exists v_{222} (v_{211} = ((v_{222} \cdot S((S(v_{217}) \cdot v_{212}))) + v_{219}))) \wedge ((\exists v_{223} ((v_{220} + S(v_{223})) =$
 $S((S(v_{217}) \cdot v_{212}))) \wedge \exists v_{224} (v_{211} = ((v_{224} \cdot S((S(v_{217}) \cdot v_{212}))) + v_{220}))) \wedge (v_{220} = (v_{219} + v_{219})))))) \wedge ((v_{197} = ((v_{209} \cdot (v_{208} \cdot 1$
 15292150460684697
 $6)) + (v_{200} \cdot v_{208} + v_{210}))) \wedge \exists v_{225} ((v_{210} + S(v_{225})) = v_{208})) \wedge (\exists v_{226} \exists v_{227} \exists v_{228} (\exists v_{229} \exists v_{230} ((\exists v_{231} ((1 + S(v_{231})) =$
 $S(v_{230})) \wedge \exists v_{232} (v_{229} = ((v_{232} \cdot S(v_{230})) + 1))) \wedge ((\exists v_{233} ((v_{226} + S(v_{233})) = S((S(v_{200} \cdot 6$
 $0)) \cdot v_{230}))) \wedge \exists v_{234} (v_{229} = ((v_{234} \cdot S((S(v_{200} \cdot 60)) \cdot v_{230}))) + v_{226}))) \wedge \forall v_{235} (\exists v_{236} ((v_{235} + S(v_{236})) = (v_{200} \cdot 6$
 $0)) \rightarrow \exists v_{237} \exists v_{238} ((\exists v_{239} ((v_{237} + S(v_{239})) = S((S(v_{235}) \cdot v_{230}))) \wedge \exists v_{240} (v_{229} =$
 $((v_{240} \cdot S((S(v_{235}) \cdot v_{230}))) + v_{237}))) \wedge ((\exists v_{241} ((v_{238} + S(v_{241})) = S((S(v_{235}) \cdot v_{230}))) \wedge \exists v_{242} (v_{229} =$
 $((v_{242} \cdot S((S(v_{235}) \cdot v_{230}))) + v_{238}))) \wedge (v_{238} = (v_{237} + v_{237})))))) \wedge ((v_{197} = ((v_{227} \cdot (v_{226} \cdot 115292150460684697$
 $6)) + (v_{202} \cdot v_{226} + v_{228}))) \wedge \exists v_{243} ((v_{228} + S(v_{243})) = v_{226}))) \wedge ((v_{202} = (720575940379279360 + ((v_{203} \cdot 26843545$
 $6) + v_{204}))) \wedge (\exists v_{244} \exists v_{245} \exists v_{246} (\exists v_{247} \exists v_{248} ((\exists v_{249} ((1 + S(v_{249})) = S(v_{248})) \wedge \exists v_{250} (v_{247} =$
 $((v_{250} \cdot S(v_{248})) + 1))) \wedge ((\exists v_{251} ((v_{244} + S(v_{251})) = S(v_{248})) \wedge \exists v_{252} (v_{247} =$
 $((v_{252} \cdot S(v_{248})) + v_{244}))) \wedge \forall v_{253} (\exists v_{254} ((v_{253} + S(v_{254})) = 0) \rightarrow \exists v_{255} \exists v_{256} ((\exists v_{257} ((v_{255} + S(v_{257})) =$
 $S((S(v_{253}) \cdot v_{248}))) \wedge \exists v_{258} (v_{247} = ((v_{258} \cdot S((S(v_{253}) \cdot v_{248}))) + v_{255}))) \wedge ((\exists v_{259} ((v_{256} + S(v_{259})) =$
 $S((S(v_{253}) \cdot v_{248}))) \wedge \exists v_{260} (v_{247} = ((v_{260} \cdot S((S(v_{253}) \cdot v_{248}))) + v_{256}))) \wedge (v_{256} = (v_{255} + v_{255})))))) \wedge ((v_{203} = ((v_{245} \cdot (v_{244} \cdot 1$
 15292150460684697
 $6)) + (v_{201} \cdot v_{244} + v_{246}))) \wedge \exists v_{261} ((v_{246} + S(v_{261})) = v_{244}))) \wedge (\exists v_{262} \exists v_{263} \exists v_{264} (\exists v_{265} \exists v_{266} ((\exists v_{267} ((1 + S(v_{267})) =$
 $S(v_{266})) \wedge \exists v_{268} (v_{265} = ((v_{268} \cdot S(v_{266})) + 1))) \wedge ((\exists v_{269} ((v_{262} + S(v_{269})) = S((S(v_{201} \cdot 6$
 $0)) \cdot v_{266}))) \wedge \exists v_{270} (v_{265} = ((v_{270} \cdot S((S(v_{201} \cdot 60)) \cdot v_{266}))) + v_{262}))) \wedge \forall v_{271} (\exists v_{272} ((v_{271} + S(v_{272})) = (v_{201} \cdot 6$
 $0)) \rightarrow \exists v_{273} \exists v_{274} ((\exists v_{275} ((v_{273} + S(v_{275})) = S((S(v_{271}) \cdot v_{266}))) \wedge \exists v_{276} (v_{265} =$
 $((v_{276} \cdot S((S(v_{271}) \cdot v_{266}))) + v_{273}))) \wedge ((\exists v_{277} ((v_{274} + S(v_{277})) = S((S(v_{271}) \cdot v_{266}))) \wedge \exists v_{278} (v_{265} =$
 $((v_{278} \cdot S((S(v_{271}) \cdot v_{266}))) + v_{274}))) \wedge (v_{274} = (v_{273} + v_{273})))))) \wedge ((v_{203} = ((v_{263} \cdot (v_{262} \cdot 115292150460684697$
 $6)) + (v_{203} \cdot v_{262} + v_{264}))) \wedge \exists v_{279} ((v_{264} + S(v_{279})) = v_{262}))) \wedge ((v_{203} = (792633534417207296 + ((v_{206} \cdot 26843545$
 $6) + v_{205}))) \wedge \exists v_{280} \exists v_{281} \exists v_{282} (\exists v_{283} \exists v_{284} \exists v_{285} (\exists v_{286} \exists v_{287} ((\exists v_{288} ((1 + S(v_{288})) = S(v_{287})) \wedge \exists v_{289} (v_{286} =$
 $((v_{289} \cdot S(v_{287})) + 1))) \wedge ((\exists v_{290} ((v_{283} + S(v_{290})) = S(v_{287})) \wedge \exists v_{291} (v_{286} =$
 $((v_{291} \cdot S(v_{287})) + v_{283}))) \wedge \forall v_{292} (\exists v_{293} ((v_{292} + S(v_{293})) = 0) \rightarrow \exists v_{294} \exists v_{295} ((\exists v_{296} ((v_{294} + S(v_{296})) =$
 $S((S(v_{292}) \cdot v_{287}))) \wedge \exists v_{297} (v_{286} = ((v_{297} \cdot S((S(v_{292}) \cdot v_{287}))) + v_{294}))) \wedge ((\exists v_{298} ((v_{295} + S(v_{298})) =$
 $S((S(v_{292}) \cdot v_{287}))) \wedge \exists v_{299} (v_{286} = ((v_{299} \cdot S((S(v_{292}) \cdot v_{287}))) + v_{295}))) \wedge (v_{295} = (v_{294} + v_{294})))))) \wedge ((v_{205} = ((v_{284} \cdot (v_{283} \cdot 1$
 15292150460684697
 $6)) + (v_{280} \cdot v_{283} + v_{285}))) \wedge \exists v_{300} ((v_{285} + S(v_{300})) = v_{283}))) \wedge (\exists v_{301} \exists v_{302} \exists v_{303} (\exists v_{304} \exists v_{305} ((\exists v_{306} ((1 + S(v_{306})) =$
 $S(v_{305})) \wedge \exists v_{307} (v_{304} = ((v_{307} \cdot S(v_{305})) + 1))) \wedge ((\exists v_{308} ((v_{301} + S(v_{308})) = S(v_{305})) \wedge \exists v_{309} (v_{304} =$
 $((v_{309} \cdot S(v_{305})) + v_{301}))) \wedge \forall v_{310} (\exists v_{311} ((v_{310} + S(v_{311})) = 0) \rightarrow \exists v_{312} \exists v_{313} ((\exists v_{314} ((v_{312} + S(v_{314})) =$
 $S((S(v_{310}) \cdot v_{305}))) \wedge \exists v_{315} (v_{304} = ((v_{315} \cdot S((S(v_{310}) \cdot v_{305}))) + v_{312}))) \wedge ((\exists v_{316} ((v_{313} + S(v_{316})) =$
 $S((S(v_{310}) \cdot v_{305}))) \wedge \exists v_{317} (v_{304} = ((v_{317} \cdot S((S(v_{310}) \cdot v_{305}))) + v_{313}))) \wedge (v_{313} = (v_{312} + v_{312})))))) \wedge ((v_{204} = ((v_{302} \cdot (v_{301} \cdot 1$
 $152921504606846976)) + ((v_{280} \cdot v_{301} + v_{303}))) \wedge \exists v_{318} ((v_{303} + S(v_{318})) = v_{301}))) \wedge (\exists v_{319} \exists v_{320} ((\exists v_{321} ((1 + S(v_{321})) =$
 $S(v_{320})) \wedge \exists v_{322} (v_{319} = ((v_{322} \cdot S(v_{320})) + 1))) \wedge ((\exists v_{323} ((v_{281} + S(v_{323})) = S((29 \cdot v_{320}))) \wedge \exists v_{324} (v_{319} = ((v_{324} \cdot S((2$
 $9 \cdot v_{320}))) + v_{281}))) \wedge \forall v_{325} (\exists v_{326} ((v_{325} + S(v_{326})) = 2$
 $8) \rightarrow \exists v_{327} \exists v_{328} ((\exists v_{329} ((v_{327} + S(v_{329})) = S((S(v_{325}) \cdot v_{320}))) \wedge \exists v_{330} (v_{319} =$
 $((v_{330} \cdot S((S(v_{325}) \cdot v_{320}))) + v_{327}))) \wedge ((\exists v_{331} ((v_{328} + S(v_{331})) = S((S(v_{325}) \cdot v_{320}))) \wedge \exists v_{332} (v_{319} =$
 $((v_{332} \cdot S((S(v_{325}) \cdot v_{320}))) + v_{328}))) \wedge (v_{328} = (v_{327} + v_{327})))))) \wedge ((\exists v_{333} \exists v_{334} ((\exists v_{335} ((1 + S(v_{335})) =$
 $S(v_{334})) \wedge \exists v_{336} (v_{333} = ((v_{336} \cdot S(v_{334})) + 1))) \wedge ((\exists v_{337} ((v_{282} + S(v_{337})) = S((57 \cdot v_{334}))) \wedge \exists v_{338} (v_{333} = ((v_{338} \cdot S((5$
 $7 \cdot v_{334}))) + v_{282}))) \wedge \forall v_{339} (\exists v_{340} ((v_{339} + S(v_{340})) = 5$
 $6) \rightarrow \exists v_{341} \exists v_{342} ((\exists v_{343} ((v_{341} + S(v_{343})) = S((S(v_{339}) \cdot v_{334}))) \wedge \exists v_{344} (v_{333} =$
 $((v_{344} \cdot S((S(v_{339}) \cdot v_{334}))) + v_{341}))) \wedge ((\exists v_{345} ((v_{342} + S(v_{345})) = S((S(v_{339}) \cdot v_{334}))) \wedge \exists v_{346} (v_{333} =$
 $((v_{346} \cdot S((S(v_{339}) \cdot v_{334}))) + v_{342}))) \wedge (v_{342} = (v_{341} + v_{341})))))) \wedge \forall v_{347} ((\exists v_{348} ((v_{347} + S(v_{348})) = v_{280}) \vee (v_{347} = v_{280})) \rightarrow$
 $((\exists v_{349} (S(v_{349}) = v_{347}) \wedge \exists v_{350} ((v_{347} + S(v_{350})) = S(v_{280}))) \rightarrow \exists v_{351} \exists v_{352} (\exists v_{353} \exists v_{354} \exists v_{355} (\exists v_{356} \exists v_{357} ((\exists v_{358} ((1 + S(v_{358})) =$

$$\begin{aligned}
& S(v_{357})) \wedge \exists v_{359}(v_{356} = ((v_{359} \cdot S(v_{357})) + 1)) \wedge ((\exists v_{360}((v_{353} + S(v_{360})) = S((S((v_{347} \cdot 6 \\
& 0)) \cdot v_{357}))) \wedge \exists v_{361}(v_{356} = ((v_{361} \cdot S((S((v_{347} \cdot 60)) \cdot v_{357}))) + v_{353}))) \wedge \forall v_{362}(\exists v_{363}((v_{362} + S(v_{363})) = (v_{347} \cdot 6 \\
& 0)) \rightarrow \exists v_{364} \exists v_{365}((\exists v_{366}((v_{364} + S(v_{366})) = S((S(v_{362}) \cdot v_{357}))) \wedge \exists v_{367}(v_{356} = \\
& ((v_{367} \cdot S((S(v_{362}) \cdot v_{357}))) + v_{364}))) \wedge ((\exists v_{368}((v_{365} + S(v_{368})) = S((S(v_{362}) \cdot v_{357}))) \wedge \exists v_{369}(v_{356} = \\
& ((v_{369} \cdot S((S(v_{362}) \cdot v_{357}))) + v_{365}))) \wedge (v_{365} = (v_{364} + v_{364})))))) \wedge ((v_{205} = ((v_{354} \cdot (v_{353} \cdot 115292150460684697 \\
& 6)) + ((v_{351} \cdot v_{353}) + v_{355}))) \wedge \exists v_{370}((v_{355} + S(v_{370})) = v_{353})) \wedge (\exists v_{371} \exists v_{372} \exists v_{373}(\exists v_{374} \exists v_{375}((\exists v_{376}((1 + S(v_{376})) = \\
& S(v_{375})) \wedge \exists v_{377}(v_{374} = ((v_{377} \cdot S(v_{375})) + 1)) \wedge ((\exists v_{378}((v_{371} + S(v_{378})) = S((S((v_{347} \cdot 6 \\
& 0)) \cdot v_{375}))) \wedge \exists v_{379}(v_{374} = ((v_{379} \cdot S((S((v_{347} \cdot 60)) \cdot v_{375}))) + v_{371}))) \wedge \forall v_{380}(\exists v_{381}((v_{380} + S(v_{381})) = (v_{347} \cdot 6 \\
& 0)) \rightarrow \exists v_{382} \exists v_{383}((\exists v_{384}((v_{382} + S(v_{384})) = S((S(v_{380}) \cdot v_{375}))) \wedge \exists v_{385}(v_{374} = \\
& ((v_{385} \cdot S((S(v_{380}) \cdot v_{375}))) + v_{382}))) \wedge ((\exists v_{386}((v_{383} + S(v_{386})) = S((S(v_{380}) \cdot v_{375}))) \wedge \exists v_{387}(v_{374} = \\
& ((v_{387} \cdot S((S(v_{380}) \cdot v_{375}))) + v_{383}))) \wedge (v_{383} = (v_{382} + v_{382})))))) \wedge ((v_{204} = ((v_{372} \cdot (v_{371} \cdot 115292150460684697 \\
& 6)) + ((v_{352} \cdot v_{371}) + v_{373}))) \wedge \exists v_{388}((v_{373} + S(v_{388})) = v_{371})) \wedge \exists v_{389} \exists v_{390} \exists v_{391} \exists v_{392}((v_{351} = \\
& ((v_{389} \cdot v_{282}) + ((v_{390} \cdot v_{281}) + v_{392}))) \wedge ((v_{352} = ((v_{389} \cdot v_{282}) + ((v_{391} \cdot v_{281}) + v_{392}))) \wedge (((v_{389} = 2) \wedge (v_{390} = v_{206})) \rightarrow \\
& (v_{391} = v_{207})) \wedge \neg((v_{389} = 2) \wedge (v_{390} = v_{206})) \rightarrow (v_{391} = v_{390})))))) \vee \\
& (\exists v_{393} \exists v_{394} \exists v_{395} \exists v_{396} \exists v_{397} \exists v_{398} \exists v_{399} \exists v_{400} \exists v_{401} \exists v_{402}(\exists v_{403} \exists v_{404} \exists v_{405}(\exists v_{406} \exists v_{407}((\exists v_{408}((1 + S(v_{408})) = \\
& S(v_{407})) \wedge \exists v_{409}(v_{406} = ((v_{409} \cdot S(v_{407})) + 1)) \wedge ((\exists v_{410}((v_{403} + S(v_{410})) = S(v_{407})) \wedge \exists v_{411}(v_{406} = \\
& ((v_{411} \cdot S(v_{407})) + v_{403}))) \wedge \forall v_{412}(\exists v_{413}((v_{412} + S(v_{413})) = 0) \rightarrow \exists v_{414} \exists v_{415}((\exists v_{416}((v_{414} + S(v_{416})) = \\
& S((S(v_{412}) \cdot v_{407}))) \wedge \exists v_{417}(v_{406} = ((v_{417} \cdot S((S(v_{412}) \cdot v_{407}))) + v_{414}))) \wedge ((\exists v_{418}((v_{415} + S(v_{418})) = \\
& S((S(v_{412}) \cdot v_{407}))) \wedge \exists v_{419}(v_{406} = ((v_{419} \cdot S((S(v_{412}) \cdot v_{407}))) + v_{415}))) \wedge (v_{415} = (v_{414} + v_{414})))))) \wedge ((v_{197} = ((v_{404} \cdot (v_{403} \cdot 1 \\
& 15292150460684697 \\
& 6)) + ((v_{393} \cdot v_{403}) + v_{405}))) \wedge \exists v_{420}((v_{405} + S(v_{420})) = v_{403})) \wedge (\exists v_{421} \exists v_{422} \exists v_{423}(\exists v_{424} \exists v_{425}((\exists v_{426}((1 + S(v_{426})) = \\
& S(v_{425})) \wedge \exists v_{427}(v_{424} = ((v_{427} \cdot S(v_{425})) + 1)) \wedge ((\exists v_{428}((v_{421} + S(v_{428})) = S((S(v_{393} \cdot 6 \\
& 0)) \cdot v_{425}))) \wedge \exists v_{429}(v_{424} = ((v_{429} \cdot S((S(v_{393} \cdot 60)) \cdot v_{425}))) + v_{421})) \wedge \forall v_{430}(\exists v_{431}((v_{430} + S(v_{431})) = (v_{393} \cdot 6 \\
& 0)) \rightarrow \exists v_{432} \exists v_{433}((\exists v_{434}((v_{432} + S(v_{434})) = S((S(v_{430}) \cdot v_{425}))) \wedge \exists v_{435}(v_{424} = \\
& ((v_{435} \cdot S((S(v_{430}) \cdot v_{425}))) + v_{432}))) \wedge ((\exists v_{436}((v_{433} + S(v_{436})) = S((S(v_{430}) \cdot v_{425}))) \wedge \exists v_{437}(v_{424} = \\
& ((v_{437} \cdot S((S(v_{430}) \cdot v_{425}))) + v_{433}))) \wedge (v_{433} = (v_{432} + v_{432})))))) \wedge ((v_{197} = ((v_{422} \cdot (v_{421} \cdot 115292150460684697 \\
& 6)) + ((v_{394} \cdot v_{421}) + v_{423}))) \wedge \exists v_{438}((v_{423} + S(v_{438})) = v_{421})) \wedge ((v_{394} = (720575940379279360 + ((v_{395} \cdot 26843545 \\
& 6) + v_{396}))) \wedge (\exists v_{439} \exists v_{440} \exists v_{441}(\exists v_{442} \exists v_{443}(\exists v_{444}((1 + S(v_{444})) = S(v_{443})) \wedge \exists v_{445}(v_{442} = \\
& ((v_{445} \cdot S(v_{443})) + 1)) \wedge ((\exists v_{446}((v_{439} + S(v_{446})) = S(v_{443})) \wedge \exists v_{447}(v_{442} = \\
& ((v_{447} \cdot S(v_{443})) + v_{439}))) \wedge \forall v_{448}(\exists v_{449}((v_{448} + S(v_{449})) = 0) \rightarrow \exists v_{450} \exists v_{451}((\exists v_{452}((v_{450} + S(v_{452})) = \\
& S((S(v_{448}) \cdot v_{443}))) \wedge \exists v_{453}(v_{442} = ((v_{453} \cdot S((S(v_{448}) \cdot v_{443}))) + v_{450}))) \wedge ((\exists v_{454}((v_{451} + S(v_{454})) = \\
& S((S(v_{448}) \cdot v_{443}))) \wedge \exists v_{455}(v_{442} = ((v_{455} \cdot S((S(v_{448}) \cdot v_{443}))) + v_{451}))) \wedge (v_{451} = (v_{450} + v_{450})))))) \wedge ((v_{395} = ((v_{440} \cdot (v_{439} \cdot 1 \\
& 152921504606846976) + v_{441})) \wedge \exists v_{456}((v_{441} + S(v_{456})) = v_{439})) \wedge (\exists v_{457} \exists v_{458} \exists v_{459}(\exists v_{460} \exists v_{461}((\exists v_{462}((1 + S(v_{462})) = \\
& S(v_{461})) \wedge \exists v_{463}(v_{460} = ((v_{463} \cdot S(v_{461})) + 1)) \wedge ((\exists v_{464}((v_{457} + S(v_{464})) = S((61 \cdot v_{461}))) \wedge \exists v_{465}(v_{460} = ((v_{465} \cdot S((6 \\
& 1 \cdot v_{461}))) + v_{457}))) \wedge \forall v_{466}(\exists v_{467}((v_{466} + S(v_{467})) = 6 \\
& 0) \rightarrow \exists v_{468} \exists v_{469}((\exists v_{470}((v_{468} + S(v_{470})) = S((S(v_{466}) \cdot v_{461}))) \wedge \exists v_{471}(v_{460} = \\
& ((v_{471} \cdot S((S(v_{466}) \cdot v_{461}))) + v_{468}))) \wedge ((\exists v_{472}((v_{469} + S(v_{472})) = S((S(v_{466}) \cdot v_{461}))) \wedge \exists v_{473}(v_{460} = \\
& ((v_{473} \cdot S((S(v_{466}) \cdot v_{461}))) + v_{469}))) \wedge (v_{469} = (v_{468} + v_{468})))))) \wedge ((v_{395} = ((v_{458} \cdot (v_{457} \cdot 115292150460684697 \\
& 6)) + ((v_{397} \cdot v_{457}) + v_{459}))) \wedge \exists v_{474}((v_{459} + S(v_{474})) = v_{457})) \wedge ((v_{397} = (360287970189639680 + ((v_{398} \cdot 26843545 \\
& 6) + v_{399}))) \wedge (\exists v_{475} \exists v_{476} \exists v_{477}(\exists v_{478} \exists v_{479}((\exists v_{480}((1 + S(v_{480})) = S(v_{479})) \wedge \exists v_{481}(v_{478} = \\
& ((v_{481} \cdot S(v_{479})) + 1)) \wedge ((\exists v_{482}((v_{475} + S(v_{482})) = S((61 \cdot v_{479}))) \wedge \exists v_{483}(v_{478} = ((v_{483} \cdot S((6 \\
& 1 \cdot v_{479}))) + v_{475}))) \wedge \forall v_{484}(\exists v_{485}((v_{484} + S(v_{485})) = 6 \\
& 0) \rightarrow \exists v_{486} \exists v_{487}((\exists v_{488}((v_{486} + S(v_{488})) = S((S(v_{484}) \cdot v_{479}))) \wedge \exists v_{489}(v_{478} = \\
& ((v_{489} \cdot S((S(v_{484}) \cdot v_{479}))) + v_{486}))) \wedge ((\exists v_{490}((v_{487} + S(v_{490})) = S((S(v_{484}) \cdot v_{479}))) \wedge \exists v_{491}(v_{478} = \\
& ((v_{491} \cdot S((S(v_{484}) \cdot v_{479}))) + v_{487}))) \wedge (v_{487} = (v_{486} + v_{486})))))) \wedge ((v_{396} = ((v_{476} \cdot (v_{475} \cdot 115292150460684697 \\
& 6)) + ((v_{400} \cdot v_{475}) + v_{477}))) \wedge \exists v_{492}((v_{477} + S(v_{492})) = v_{475})) \wedge ((v_{400} = (720575940379279360 + ((v_{401} \cdot 26843545 \\
& 6) + v_{402}))) \wedge \exists v_{493} \exists v_{494} \exists v_{495}(\exists v_{496} \exists v_{497} \exists v_{498}(\exists v_{499} \exists v_{500}((\exists v_{501}((1 + S(v_{501})) = S(v_{500})) \wedge \exists v_{502}(v_{499} = \\
& ((v_{502} \cdot S(v_{500})) + 1)) \wedge ((\exists v_{503}((v_{496} + S(v_{503})) = S(v_{500})) \wedge \exists v_{504}(v_{499} = \\
& ((v_{504} \cdot S(v_{500})) + v_{496}))) \wedge \forall v_{505}(\exists v_{506}((v_{505} + S(v_{506})) = 0) \rightarrow \exists v_{507} \exists v_{508}((\exists v_{509}((v_{507} + S(v_{509})) = \\
& S((S(v_{505}) \cdot v_{500}))) \wedge \exists v_{510}(v_{499} = ((v_{510} \cdot S((S(v_{505}) \cdot v_{500}))) + v_{507}))) \wedge ((\exists v_{511}((v_{508} + S(v_{511})) = \\
& S((S(v_{505}) \cdot v_{500}))) \wedge \exists v_{512}(v_{499} = ((v_{512} \cdot S((S(v_{505}) \cdot v_{500}))) + v_{508}))) \wedge (v_{508} = (v_{507} + v_{507})))))) \wedge ((v_{401} = ((v_{497} \cdot (v_{496} \cdot 1 \\
& 15292150460684697 \\
& 6)) + ((v_{493} \cdot v_{496}) + v_{498}))) \wedge \exists v_{513}((v_{498} + S(v_{513})) = v_{496})) \wedge (\exists v_{514} \exists v_{515} \exists v_{516}(\exists v_{517} \exists v_{518}((\exists v_{519}((1 + S(v_{519})) = \\
& S(v_{518})) \wedge \exists v_{520}(v_{517} = ((v_{520} \cdot S(v_{518})) + 1)) \wedge ((\exists v_{521}((v_{514} + S(v_{521})) = S(v_{518})) \wedge \exists v_{522}(v_{517} = \\
& ((v_{522} \cdot S(v_{518})) + v_{514}))) \wedge \forall v_{523}(\exists v_{524}((v_{523} + S(v_{524})) = 0) \rightarrow \exists v_{525} \exists v_{526}((\exists v_{527}((v_{525} + S(v_{527})) = \\
& S((S(v_{523}) \cdot v_{518}))) \wedge \exists v_{528}(v_{517} = ((v_{528} \cdot S((S(v_{523}) \cdot v_{518}))) + v_{525}))) \wedge ((\exists v_{529}((v_{526} + S(v_{529})) = \\
& S((S(v_{523}) \cdot v_{518}))) \wedge \exists v_{530}(v_{517} = ((v_{530} \cdot S((S(v_{523}) \cdot v_{518}))) + v_{526}))) \wedge (v_{526} = (v_{525} + v_{525})))))) \wedge ((v_{402} = ((v_{515} \cdot (v_{514} \cdot 1 \\
& 152921504606846976) + ((v_{493} \cdot v_{514}) + v_{516}))) \wedge \exists v_{531}((v_{516} + S(v_{531})) = v_{514})) \wedge (\exists v_{532} \exists v_{533}((\exists v_{534}((1 + S(v_{534})) = \\
& S(v_{533})) \wedge \exists v_{535}(v_{532} = ((v_{535} \cdot S(v_{533})) + 1)) \wedge ((\exists v_{536}((v_{494} + S(v_{536})) = S((29 \cdot v_{533}))) \wedge \exists v_{537}(v_{532} = ((v_{537} \cdot S((2 \\
& 9 \cdot v_{533}))) + v_{494}))) \wedge \forall v_{538}(\exists v_{539}((v_{538} + S(v_{539})) = 2 \\
& 8) \rightarrow \exists v_{540} \exists v_{541}((\exists v_{542}((v_{540} + S(v_{542})) = S((S(v_{538}) \cdot v_{533}))) \wedge \exists v_{543}(v_{532} = \\
& ((v_{543} \cdot S((S(v_{538}) \cdot v_{533}))) + v_{540}))) \wedge ((\exists v_{544}((v_{541} + S(v_{544})) = S((S(v_{538}) \cdot v_{533}))) \wedge \exists v_{545}(v_{532} = \\
& ((v_{545} \cdot S((S(v_{538}) \cdot v_{533}))) + v_{541}))) \wedge (v_{541} = (v_{540} + v_{540})))))) \wedge (\exists v_{546} \exists v_{547}((\exists v_{548}((1 + S(v_{548})) = \\
& S(v_{547})) \wedge \exists v_{549}(v_{546} = ((v_{549} \cdot S(v_{547})) + 1)) \wedge ((\exists v_{550}((v_{495} + S(v_{550})) = S((57 \cdot v_{547}))) \wedge \exists v_{551}(v_{546} = ((v_{551} \cdot S((5 \\
& 7 \cdot v_{547}))) + v_{495}))) \wedge \forall v_{552}(\exists v_{553}((v_{552} + S(v_{553})) = 5 \\
& 6) \rightarrow \exists v_{554} \exists v_{555}((\exists v_{556}((v_{554} + S(v_{556})) = S((S(v_{552}) \cdot v_{547}))) \wedge \exists v_{557}(v_{546} = \\
& ((v_{557} \cdot S((S(v_{552}) \cdot v_{547}))) + v_{554}))) \wedge ((\exists v_{558}((v_{555} + S(v_{558})) = S((S(v_{552}) \cdot v_{547}))) \wedge \exists v_{559}(v_{546} = \\
& ((v_{559} \cdot S((S(v_{552}) \cdot v_{547}))) + v_{555}))) \wedge (v_{555} = (v_{554} + v_{554})))))) \wedge \forall v_{560}(\exists v_{561}((v_{560} + S(v_{561})) = v_{493}) \vee (v_{560} = v_{493})) \rightarrow
\end{aligned}$$

$$\begin{aligned}
& ((\exists v_{562}(S(v_{562}) = v_{560}) \wedge \exists v_{563}((v_{560} + S(v_{563})) = S(v_{493}))) \rightarrow \exists v_{564} \exists v_{565} (\exists v_{566} \exists v_{567} \exists v_{568} (\exists v_{569} \exists v_{570} ((\exists v_{571}((1 + S(v_{571})) = \\
& S(v_{570})) \wedge \exists v_{572}(v_{569} = ((v_{572} \cdot S(v_{570})) + 1)) \wedge ((\exists v_{573}((v_{566} + S(v_{573})) = S((S((v_{560} \cdot 6 \\
& 0)) \cdot v_{570}))) \wedge \exists v_{574}(v_{569} = ((v_{574} \cdot S((S((v_{560} \cdot 60)) \cdot v_{570}))) + v_{566}))) \wedge \forall v_{575} (\exists v_{576} (v_{575} + S(v_{576})) = (v_{560} \cdot 6 \\
& 0)) \rightarrow \exists v_{577} \exists v_{578} ((\exists v_{579} ((v_{577} + S(v_{579})) = S((S(v_{575}) \cdot v_{570}))) \wedge \exists v_{580}(v_{569} = \\
& ((v_{580} \cdot S((S(v_{575}) \cdot v_{570}))) + v_{577}))) \wedge ((\exists v_{581} (v_{578} + S(v_{581})) = S((S(S(v_{575})) \cdot v_{570}))) \wedge \exists v_{582}(v_{569} = \\
& ((v_{582} \cdot S((S(S(v_{575})) \cdot v_{570}))) + v_{578}))) \wedge (v_{578} = (v_{577} + v_{577})))))) \wedge ((v_{401} = ((v_{567} \cdot (v_{566} \cdot 115292150460684697 \\
& 6)) + ((v_{564} \cdot v_{566}) + v_{568}))) \wedge \exists v_{583}((v_{568} + S(v_{583})) = v_{566})) \wedge (\exists v_{584} \exists v_{585} \exists v_{586} (\exists v_{587} \exists v_{588} ((\exists v_{589}((1 + S(v_{589})) = \\
& S(v_{588})) \wedge \exists v_{590}(v_{587} = ((v_{590} \cdot S(v_{588})) + 1)) \wedge ((\exists v_{591}((v_{584} + S(v_{591})) = S((S((v_{560} \cdot 6 \\
& 0)) \cdot v_{588}))) \wedge \exists v_{592}(v_{587} = ((v_{592} \cdot S((S((v_{560} \cdot 60)) \cdot v_{588}))) + v_{584}))) \wedge \forall v_{593} (\exists v_{594} (v_{593} + S(v_{594})) = (v_{560} \cdot 6 \\
& 0)) \rightarrow \exists v_{595} \exists v_{596} ((\exists v_{597} ((v_{595} + S(v_{597})) = S((S(v_{593}) \cdot v_{588}))) \wedge \exists v_{598}(v_{587} = \\
& ((v_{598} \cdot S((S(v_{593}) \cdot v_{588}))) + v_{595}))) \wedge ((\exists v_{599} ((v_{596} + S(v_{599})) = S((S(S(v_{593})) \cdot v_{588}))) \wedge \exists v_{600}(v_{587} = \\
& ((v_{600} \cdot S((S(S(v_{593})) \cdot v_{588}))) + v_{596}))) \wedge (v_{596} = (v_{595} + v_{595})))))) \wedge ((v_{402} = ((v_{585} \cdot (v_{584} \cdot 115292150460684697 \\
& 6)) + ((v_{565} \cdot v_{584}) + v_{586}))) \wedge \exists v_{601}((v_{586} + S(v_{601})) = v_{584})) \wedge \exists v_{602} \exists v_{603} \exists v_{604} \exists v_{605} ((v_{564} = \\
& ((v_{602} \cdot v_{495}) + ((v_{603} \cdot v_{494}) + v_{605}))) \wedge ((v_{605} = ((v_{602} \cdot v_{495}) + ((v_{604} \cdot v_{494}) + v_{605}))) \wedge (((v_{602} = 2) \wedge (v_{603} = v_{398})) \rightarrow \\
& (v_{604} = v_{399})) \wedge (\neg((v_{602} = 2) \wedge (v_{603} = v_{398})) \rightarrow (v_{604} = v_{603})))))) \vee \\
& \exists v_{606} \exists v_{607} \exists v_{608} \exists v_{609} \exists v_{610} \exists v_{611} \exists v_{612} \exists v_{613} \exists v_{614} \exists v_{615} \exists v_{616} \exists v_{617} \exists v_{618} \exists v_{619} \exists v_{620} (\exists v_{621} \exists v_{622} ((\exists v_{623}((1 + S(v_{623})) = \\
& S(v_{622})) \wedge \exists v_{624}(v_{621} = ((v_{624} \cdot S(v_{622})) + 1)) \wedge ((\exists v_{625}((v_{618} + S(v_{625})) = S(v_{622})) \wedge \exists v_{626}(v_{621} = \\
& ((v_{626} \cdot S(v_{622})) + v_{618}))) \wedge \forall v_{627} (\exists v_{628} ((v_{627} + S(v_{628})) = 0) \rightarrow \exists v_{629} \exists v_{630} ((\exists v_{631} ((v_{629} + S(v_{631})) = \\
& S((S(v_{627}) \cdot v_{622}))) \wedge \exists v_{632}(v_{621} = ((v_{632} \cdot S((S(v_{627}) \cdot v_{622}))) + v_{629}))) \wedge ((\exists v_{633} (v_{630} + S(v_{633})) = \\
& S((S(S(v_{627}) \cdot v_{622}))) \wedge \exists v_{634}(v_{621} = ((v_{634} \cdot S((S(S(v_{627}) \cdot v_{622}))) + v_{630}))) \wedge (v_{630} = (v_{629} + v_{629})))))) \wedge ((v_{197} = ((v_{619} \cdot (v_{618} \cdot 1 \\
& 15292150460684697 \\
& 6)) + ((v_{606} \cdot v_{618}) + v_{620}))) \wedge \exists v_{635}((v_{620} + S(v_{635})) = v_{618})) \wedge (\exists v_{636} \exists v_{637} \exists v_{638} (\exists v_{639} \exists v_{640} ((\exists v_{641}((1 + S(v_{641})) = \\
& S(v_{640})) \wedge \exists v_{642}(v_{639} = ((v_{642} \cdot S(v_{640})) + 1)) \wedge ((\exists v_{643}((v_{636} + S(v_{643})) = S((S((v_{606} \cdot 6 \\
& 0)) \cdot v_{640}))) \wedge \exists v_{644}(v_{639} = ((v_{644} \cdot S((S((v_{606} \cdot 60)) \cdot v_{640}))) + v_{636}))) \wedge \forall v_{645} (\exists v_{646} (v_{645} + S(v_{646})) = (v_{606} \cdot 6 \\
& 0)) \rightarrow \exists v_{647} \exists v_{648} ((\exists v_{649} ((v_{647} + S(v_{649})) = S((S(v_{645}) \cdot v_{640}))) \wedge \exists v_{650}(v_{639} = \\
& ((v_{650} \cdot S((S(v_{645}) \cdot v_{640}))) + v_{647}))) \wedge ((\exists v_{651} ((v_{648} + S(v_{651})) = S((S(S(v_{645})) \cdot v_{640}))) \wedge \exists v_{652}(v_{639} = \\
& ((v_{652} \cdot S((S(S(v_{645})) \cdot v_{640}))) + v_{648}))) \wedge (v_{648} = (v_{647} + v_{647})))))) \wedge ((v_{197} = ((v_{637} \cdot (v_{636} \cdot 115292150460684697 \\
& 6)) + ((v_{607} \cdot v_{636}) + v_{638}))) \wedge \exists v_{653}((v_{638} + S(v_{653})) = v_{636})) \wedge ((v_{607} = (720575940379279360 + ((v_{608} \cdot 26843545 \\
& 6) + v_{609}))) \wedge (\exists v_{654} \exists v_{655} \exists v_{656} (\exists v_{657} \exists v_{658} ((\exists v_{659}((1 + S(v_{659})) = S(v_{658})) \wedge \exists v_{660}(v_{657} = \\
& ((v_{660} \cdot S(v_{658})) + 1)) \wedge ((\exists v_{661} ((v_{654} + S(v_{661})) = S((61 \cdot v_{658}))) \wedge \exists v_{662}(v_{657} = ((v_{662} \cdot S((6 \\
& 1 \cdot v_{658}))) + v_{654}))) \wedge \forall v_{663} (\exists v_{664} ((v_{663} + S(v_{664})) = 6 \\
& 0) \rightarrow \exists v_{665} \exists v_{666} ((\exists v_{667} ((v_{665} + S(v_{667})) = S((S(v_{663}) \cdot v_{658}))) \wedge \exists v_{668}(v_{657} = \\
& ((v_{668} \cdot S((S(v_{663}) \cdot v_{658}))) + v_{665}))) \wedge ((\exists v_{669} ((v_{666} + S(v_{669})) = S((S(S(v_{663})) \cdot v_{658}))) \wedge \exists v_{670}(v_{657} = \\
& ((v_{670} \cdot S((S(S(v_{663}) \cdot v_{658}))) + v_{666}))) \wedge (v_{666} = (v_{665} + v_{665})))))) \wedge ((v_{609} = ((v_{655} \cdot (v_{654} \cdot 115292150460684697 \\
& 6)) + ((v_{609} \cdot v_{654}) + v_{656}))) \wedge \exists v_{671}((v_{656} + S(v_{671})) = v_{654})) \wedge ((v_{609} = (792633534417207296 + ((v_{610} \cdot 26843545 \\
& 6) + v_{611}))) \wedge (\exists v_{672} \exists v_{673} \exists v_{674} (\exists v_{675} \exists v_{676} ((\exists v_{677}((1 + S(v_{677})) = S(v_{676})) \wedge \exists v_{678}(v_{675} = \\
& ((v_{678} \cdot S(v_{676})) + 1)) \wedge ((\exists v_{679} ((v_{672} + S(v_{679})) = S((61 \cdot v_{676}))) \wedge \exists v_{680}(v_{675} = ((v_{680} \cdot S((6 \\
& 1 \cdot v_{676}))) + v_{672}))) \wedge \forall v_{681} (\exists v_{682} ((v_{681} + S(v_{682})) = 6 \\
& 0) \rightarrow \exists v_{683} \exists v_{684} ((\exists v_{685} ((v_{683} + S(v_{685})) = S((S(v_{681}) \cdot v_{676}))) \wedge \exists v_{686}(v_{675} = \\
& ((v_{686} \cdot S((S(v_{681}) \cdot v_{676}))) + v_{683}))) \wedge ((\exists v_{687} ((v_{684} + S(v_{687})) = S((S(S(v_{681})) \cdot v_{676}))) \wedge \exists v_{688}(v_{675} = \\
& ((v_{688} \cdot S((S(S(v_{681}) \cdot v_{676}))) + v_{684}))) \wedge (v_{684} = (v_{683} + v_{683})))))) \wedge ((v_{608} = ((v_{673} \cdot (v_{672} \cdot 115292150460684697 \\
& 6)) + ((v_{608} \cdot v_{672}) + v_{674}))) \wedge \exists v_{689}((v_{674} + S(v_{689})) = v_{672})) \wedge ((v_{608} = (576460752303423488 + ((v_{612} \cdot 26843545 \\
& 6) + v_{613}))) \wedge (\exists v_{690} \exists v_{691} \exists v_{692} (\exists v_{693} \exists v_{694} ((\exists v_{695}((1 + S(v_{695})) = S(v_{694})) \wedge \exists v_{696}(v_{693} = \\
& ((v_{696} \cdot S(v_{694})) + 1)) \wedge ((\exists v_{697} ((v_{690} + S(v_{697})) = S((61 \cdot v_{694}))) \wedge \exists v_{698}(v_{693} = ((v_{698} \cdot S((6 \\
& 1 \cdot v_{694}))) + v_{690}))) \wedge \forall v_{699} (\exists v_{700} ((v_{699} + S(v_{700})) = 6 \\
& 0) \rightarrow \exists v_{701} \exists v_{702} ((\exists v_{703} ((v_{701} + S(v_{703})) = S((S(v_{699}) \cdot v_{694}))) \wedge \exists v_{704}(v_{693} = \\
& ((v_{704} \cdot S((S(v_{699}) \cdot v_{694}))) + v_{701}))) \wedge ((\exists v_{705} ((v_{702} + S(v_{705})) = S((S(S(v_{699})) \cdot v_{694}))) \wedge \exists v_{706}(v_{693} = \\
& ((v_{706} \cdot S((S(S(v_{699}) \cdot v_{694}))) + v_{702}))) \wedge (v_{702} = (v_{701} + v_{701})))))) \wedge ((v_{613} = ((v_{691} \cdot (v_{690} \cdot 115292150460684697 \\
& 6)) + ((v_{613} \cdot v_{690}) + v_{692}))) \wedge \exists v_{707}((v_{692} + S(v_{707})) = v_{690})) \wedge ((v_{613} = (792633534417207296 + ((v_{610} \cdot 26843545 \\
& 6) + v_{614}))) \wedge (\exists v_{708} \exists v_{709} \exists v_{710} (\exists v_{711} \exists v_{712} ((\exists v_{713}((1 + S(v_{713})) = S(v_{712})) \wedge \exists v_{714}(v_{711} = \\
& ((v_{714} \cdot S(v_{712})) + 1)) \wedge ((\exists v_{715} ((v_{708} + S(v_{715})) = S((61 \cdot v_{712}))) \wedge \exists v_{716}(v_{711} = ((v_{716} \cdot S((6 \\
& 1 \cdot v_{712}))) + v_{708}))) \wedge \forall v_{717} (\exists v_{718} ((v_{717} + S(v_{718})) = 6 \\
& 0) \rightarrow \exists v_{719} \exists v_{720} ((\exists v_{721} ((v_{719} + S(v_{721})) = S((S(v_{717}) \cdot v_{712}))) \wedge \exists v_{722}(v_{711} = \\
& ((v_{722} \cdot S((S(v_{717}) \cdot v_{712}))) + v_{719}))) \wedge ((\exists v_{723} ((v_{720} + S(v_{723})) = S((S(S(v_{717})) \cdot v_{712}))) \wedge \exists v_{724}(v_{711} = \\
& ((v_{724} \cdot S((S(S(v_{717}) \cdot v_{712}))) + v_{720}))) \wedge (v_{720} = (v_{719} + v_{719})))))) \wedge ((v_{614} = ((v_{709} \cdot (v_{708} \cdot 115292150460684697 \\
& 6)) + ((v_{614} \cdot v_{708}) + v_{710}))) \wedge \exists v_{725}((v_{710} + S(v_{725})) = v_{708})) \wedge ((v_{614} = (720575940379279360 + ((v_{615} \cdot 26843545 \\
& 6) + v_{616}))) \wedge ((v_{615} = v_{611}) \wedge (\exists v_{726} \exists v_{727} \exists v_{728} (\exists v_{729} \exists v_{730} \exists v_{731} (\exists v_{732} \exists v_{733} ((\exists v_{734}((1 + S(v_{734})) = S(v_{733})) \wedge \exists v_{735}(v_{732} = \\
& ((v_{735} \cdot S(v_{733})) + 1)) \wedge ((\exists v_{736} ((v_{729} + S(v_{736})) = S(v_{733})) \wedge \exists v_{737}(v_{732} = \\
& ((v_{737} \cdot S(v_{733})) + v_{729}))) \wedge \forall v_{738} (\exists v_{739} ((v_{738} + S(v_{739})) = 0) \rightarrow \exists v_{740} \exists v_{741} ((\exists v_{742} ((v_{740} + S(v_{742})) = \\
& S((S(v_{738}) \cdot v_{733}))) \wedge \exists v_{743}(v_{732} = ((v_{743} \cdot S((S(v_{738}) \cdot v_{733}))) + v_{740}))) \wedge ((\exists v_{744} (v_{741} + S(v_{744})) = \\
& S((S(S(v_{738}) \cdot v_{733}))) \wedge \exists v_{745}(v_{732} = ((v_{745} \cdot S((S(S(v_{738}) \cdot v_{733}))) + v_{741}))) \wedge (v_{741} = (v_{740} + v_{740})))))) \wedge ((v_{611} = ((v_{730} \cdot (v_{729} \cdot 1 \\
& 15292150460684697 \\
& 6)) + ((v_{726} \cdot v_{729}) + v_{731}))) \wedge \exists v_{746}((v_{731} + S(v_{746})) = v_{729})) \wedge (\exists v_{747} \exists v_{748} \exists v_{749} (\exists v_{750} \exists v_{751} ((\exists v_{752}((1 + S(v_{752})) = \\
& S(v_{751})) \wedge \exists v_{753}(v_{750} = ((v_{753} \cdot S(v_{751})) + 1)) \wedge ((\exists v_{754} ((v_{747} + S(v_{754})) = S(v_{751})) \wedge \exists v_{755}(v_{750} = \\
& ((v_{755} \cdot S(v_{751})) + v_{747}))) \wedge \forall v_{756} (\exists v_{757} ((v_{756} + S(v_{757})) = 0) \rightarrow \exists v_{758} \exists v_{759} ((\exists v_{760} ((v_{758} + S(v_{760})) = \\
& S((S(v_{756}) \cdot v_{751}))) \wedge \exists v_{761}(v_{750} = ((v_{761} \cdot S((S(v_{756}) \cdot v_{751}))) + v_{758}))) \wedge ((\exists v_{762} ((v_{759} + S(v_{762})) = \\
& S((S(S(v_{756}) \cdot v_{751}))) \wedge \exists v_{763}(v_{750} = ((v_{763} \cdot S((S(S(v_{756}) \cdot v_{751}))) + v_{759}))) \wedge (v_{759} = (v_{758} + v_{758})))))) \wedge ((v_{612} = ((v_{748} \cdot (v_{747} \cdot 1 \\
& 152921504606846976)) + ((v_{726} \cdot v_{747}) + v_{749}))) \wedge \exists v_{764}((v_{749} + S(v_{764})) = v_{747})) \wedge (\exists v_{765} \exists v_{766} ((\exists v_{767}((1 + S(v_{767})) =
\end{aligned}$$

$S(v_{766})) \wedge \exists v_{768}(v_{765} = ((v_{768} \cdot S(v_{766})) + 1)) \wedge ((\exists v_{769}((v_{727} + S(v_{769})) = S((29 \cdot v_{766}))) \wedge \exists v_{770}(v_{765} = ((v_{770} \cdot S((29 \cdot v_{766}))) + v_{727}))) \wedge \forall v_{771}(\exists v_{772}((v_{771} + S(v_{772})) = 2$
 $8) \rightarrow \exists v_{773} \exists v_{774}((\exists v_{775}((v_{773} + S(v_{775})) = S((S(v_{771}) \cdot v_{766}))) \wedge \exists v_{776}(v_{765} = ((v_{776} \cdot S((S(v_{771}) \cdot v_{766}))) + v_{773}))) \wedge ((\exists v_{777}((v_{774} + S(v_{777})) = S((S(v_{771}) \cdot v_{766}))) \wedge \exists v_{778}(v_{765} = ((v_{778} \cdot S((S(v_{771}) \cdot v_{766}))) + v_{774}))) \wedge (v_{774} = (v_{773} + v_{773})))))) \wedge (\exists v_{779} \exists v_{780}((\exists v_{781}((1 + S(v_{781})) = S(v_{780})) \wedge \exists v_{782}(v_{779} = ((v_{782} \cdot S(v_{780})) + 1)) \wedge ((\exists v_{783}((v_{728} + S(v_{783})) = S((57 \cdot v_{780}))) \wedge \exists v_{784}(v_{779} = ((v_{784} \cdot S((57 \cdot v_{780}))) + v_{728}))) \wedge \forall v_{785}(\exists v_{786}((v_{785} + S(v_{786})) = 5$
 $6) \rightarrow \exists v_{787} \exists v_{788}((\exists v_{789}((v_{787} + S(v_{789})) = S((S(v_{785}) \cdot v_{780}))) \wedge \exists v_{790}(v_{779} = ((v_{790} \cdot S((S(v_{785}) \cdot v_{780}))) + v_{787}))) \wedge ((\exists v_{791}((v_{788} + S(v_{791})) = S((S(v_{785}) \cdot v_{780}))) \wedge \exists v_{792}(v_{779} = ((v_{792} \cdot S((S(v_{785}) \cdot v_{780}))) + v_{788}))) \wedge (v_{788} = (v_{787} + v_{787})))))) \wedge \forall v_{793}((\exists v_{794}((v_{793} + S(v_{794})) = v_{726}) \vee (v_{793} = v_{726})) \rightarrow ((\exists v_{795}(S(v_{795}) = v_{793}) \wedge \exists v_{796}((v_{793} + S(v_{796})) = S(v_{726}))) \rightarrow \exists v_{797} \exists v_{798}(\exists v_{799} \exists v_{800} \exists v_{801}(\exists v_{802} \exists v_{803}((\exists v_{804}((1 + S(v_{804})) = S(v_{803})) \wedge \exists v_{805}(v_{802} = ((v_{805} \cdot S(v_{803})) + 1)) \wedge ((\exists v_{806}((v_{799} + S(v_{806})) = S((S(v_{793} \cdot 60) \cdot v_{803}))) \wedge \exists v_{807}(v_{802} = ((v_{807} \cdot S((S(v_{793} \cdot 60) \cdot v_{803}))) + v_{799}))) \wedge \forall v_{808}(\exists v_{809}((v_{808} + S(v_{809})) = (v_{793} \cdot 60) \cdot v_{803}))) \wedge \exists v_{810} \exists v_{811}((\exists v_{812}((v_{810} + S(v_{812})) = S((S(v_{808}) \cdot v_{803}))) \wedge \exists v_{813}(v_{802} = ((v_{813} \cdot S((S(v_{808}) \cdot v_{803}))) + v_{810}))) \wedge ((\exists v_{814}((v_{811} + S(v_{814})) = S((S(v_{808}) \cdot v_{803}))) \wedge \exists v_{815}(v_{802} = ((v_{815} \cdot S((S(v_{808}) \cdot v_{803}))) + v_{811}))) \wedge (v_{811} = (v_{810} + v_{810})))))) \wedge ((v_{611} = ((v_{800} \cdot (v_{799} \cdot 1152921504606846976) + (v_{797} \cdot v_{799}) + v_{801}))) \wedge \exists v_{816}((v_{801} + S(v_{816})) = v_{799}))) \wedge (\exists v_{817} \exists v_{818} \exists v_{819}(\exists v_{820} \exists v_{821}((\exists v_{822}((1 + S(v_{822})) = S(v_{821})) \wedge \exists v_{823}(v_{820} = ((v_{823} \cdot S(v_{821})) + 1)) \wedge ((\exists v_{824}((v_{817} + S(v_{824})) = S((S(v_{793} \cdot 60) \cdot v_{821}))) \wedge \exists v_{825}(v_{820} = ((v_{825} \cdot S((S(v_{793} \cdot 60) \cdot v_{821}))) + v_{817}))) \wedge \forall v_{826}(\exists v_{827}((v_{826} + S(v_{827})) = (v_{793} \cdot 60) \cdot v_{821}))) \wedge \exists v_{828} \exists v_{829}((\exists v_{830}((v_{828} + S(v_{830})) = S((S(v_{826}) \cdot v_{821}))) \wedge \exists v_{831}(v_{820} = ((v_{831} \cdot S((S(v_{826}) \cdot v_{821}))) + v_{828}))) \wedge ((\exists v_{832}((v_{829} + S(v_{832})) = S((S(v_{826}) \cdot v_{821}))) \wedge \exists v_{833}(v_{820} = ((v_{833} \cdot S((S(v_{826}) \cdot v_{821}))) + v_{829}))) \wedge (v_{829} = (v_{828} + v_{828})))))) \wedge ((v_{612} = ((v_{818} \cdot (v_{817} \cdot 1152921504606846976) + (v_{798} \cdot v_{817}) + v_{819}))) \wedge \exists v_{834}((v_{819} + S(v_{834})) = v_{817}))) \wedge \exists v_{835} \exists v_{836} \exists v_{837} \exists v_{838}((v_{797} = ((v_{835} \cdot v_{728}) + ((v_{836} \cdot v_{727}) + v_{838}))) \wedge ((\exists v_{835} = 2) \wedge (v_{836} = v_{610})) \rightarrow (v_{837} = 83076749736557242056487941267521537) \wedge (\neg((v_{835} = 2) \wedge (v_{836} = v_{610})) \rightarrow (v_{837} = v_{836})))))) \wedge ((v_{617} = (216172782113783808 + (v_{610} \cdot 268435456))) \wedge \exists v_{839} \exists v_{840} \exists v_{841}(\exists v_{842} \exists v_{843} \exists v_{844}(\exists v_{845} \exists v_{846}((\exists v_{847}((1 + S(v_{847})) = S(v_{846})) \wedge \exists v_{848}(v_{845} = ((v_{848} \cdot S(v_{846})) + 1)) \wedge ((\exists v_{849}((v_{842} + S(v_{849})) = S(v_{846})) \wedge \exists v_{850}(v_{845} = ((v_{850} \cdot S(v_{846})) + v_{842}))) \wedge \forall v_{851}(\exists v_{852}((v_{851} + S(v_{852})) = 0) \rightarrow \exists v_{853} \exists v_{854}((\exists v_{855}((v_{853} + S(v_{855})) = S((S(v_{851}) \cdot v_{846}))) \wedge \exists v_{856}(v_{845} = ((v_{856} \cdot S((S(v_{851}) \cdot v_{846}))) + v_{853}))) \wedge ((\exists v_{857}((v_{854} + S(v_{857})) = S((S(v_{851}) \cdot v_{846}))) \wedge \exists v_{858}(v_{845} = ((v_{858} \cdot S((S(v_{851}) \cdot v_{846}))) + v_{854}))) \wedge (v_{854} = (v_{853} + v_{853})))))) \wedge ((v_{611} = ((v_{843} \cdot (v_{842} \cdot 1152921504606846976) + (v_{839} \cdot v_{842}) + v_{844}))) \wedge \exists v_{859}((v_{844} + S(v_{859})) = v_{842}))) \wedge (\exists v_{860} \exists v_{861} \exists v_{862}(\exists v_{863} \exists v_{864}((\exists v_{865}((1 + S(v_{865})) = S(v_{864})) \wedge \exists v_{866}(v_{863} = ((v_{866} \cdot S(v_{864})) + 1)) \wedge ((\exists v_{867}((v_{860} + S(v_{867})) = S(v_{864})) \wedge \exists v_{868}(v_{863} = ((v_{868} \cdot S(v_{864})) + v_{860}))) \wedge \forall v_{869}(\exists v_{870}((v_{869} + S(v_{870})) = 0) \rightarrow \exists v_{871} \exists v_{872}((\exists v_{873}((v_{871} + S(v_{873})) = S((S(v_{869}) \cdot v_{864}))) \wedge \exists v_{874}(v_{863} = ((v_{874} \cdot S((S(v_{869}) \cdot v_{864}))) + v_{871}))) \wedge ((\exists v_{875}((v_{872} + S(v_{875})) = S((S(v_{869}) \cdot v_{864}))) \wedge \exists v_{876}(v_{863} = ((v_{876} \cdot S((S(v_{869}) \cdot v_{864}))) + v_{872}))) \wedge (v_{872} = (v_{871} + v_{871})))))) \wedge ((v_{616} = ((v_{861} \cdot (v_{860} \cdot 1152921504606846976) + (v_{839} \cdot v_{860}) + v_{862}))) \wedge \exists v_{877}((v_{862} + S(v_{877})) = v_{860}))) \wedge (\exists v_{878} \exists v_{879}((\exists v_{880}((1 + S(v_{880})) = S(v_{879})) \wedge \exists v_{881}(v_{878} = ((v_{881} \cdot S(v_{879})) + 1)) \wedge ((\exists v_{882}((v_{840} + S(v_{882})) = S((29 \cdot v_{879}))) \wedge \exists v_{883}(v_{878} = ((v_{883} \cdot S((29 \cdot v_{879}))) + v_{840}))) \wedge \forall v_{884}(\exists v_{885}((v_{884} + S(v_{885})) = 2$
 $8) \rightarrow \exists v_{886} \exists v_{887}((\exists v_{888}((v_{886} + S(v_{888})) = S((S(v_{884}) \cdot v_{879}))) \wedge \exists v_{889}(v_{878} = ((v_{889} \cdot S((S(v_{884}) \cdot v_{879}))) + v_{886}))) \wedge ((\exists v_{890}((v_{887} + S(v_{890})) = S((S(v_{884}) \cdot v_{879}))) \wedge \exists v_{891}(v_{878} = ((v_{891} \cdot S((S(v_{884}) \cdot v_{879}))) + v_{887}))) \wedge (v_{887} = (v_{886} + v_{886})))))) \wedge (\exists v_{892} \exists v_{893}((\exists v_{894}((1 + S(v_{894})) = S(v_{893})) \wedge \exists v_{895}(v_{892} = ((v_{895} \cdot S(v_{893})) + 1)) \wedge ((\exists v_{896}((v_{841} + S(v_{896})) = S((57 \cdot v_{893}))) \wedge \exists v_{897}(v_{892} = ((v_{897} \cdot S((57 \cdot v_{893})) + v_{841}))) \wedge \forall v_{898}(\exists v_{899}((v_{898} + S(v_{899})) = 5$
 $6) \rightarrow \exists v_{900} \exists v_{901}((\exists v_{902}((v_{900} + S(v_{902})) = S((S(v_{898}) \cdot v_{893}))) \wedge \exists v_{903}(v_{892} = ((v_{903} \cdot S((S(v_{898}) \cdot v_{893}))) + v_{900}))) \wedge ((\exists v_{904}((v_{901} + S(v_{904})) = S((S(v_{898}) \cdot v_{893}))) \wedge \exists v_{905}(v_{892} = ((v_{905} \cdot S((S(v_{898}) \cdot v_{893}))) + v_{901}))) \wedge (v_{901} = (v_{900} + v_{900})))))) \wedge \forall v_{906}((\exists v_{907}((v_{906} + S(v_{907})) = v_{839}) \vee (v_{906} = v_{839})) \rightarrow ((\exists v_{908}(S(v_{908}) = v_{906}) \wedge \exists v_{909}((v_{906} + S(v_{909})) = S(v_{839}))) \rightarrow \exists v_{910} \exists v_{911}(\exists v_{912} \exists v_{913} \exists v_{914}(\exists v_{915} \exists v_{916}((\exists v_{917}((1 + S(v_{917})) = S(v_{916})) \wedge \exists v_{918}(v_{915} = ((v_{918} \cdot S(v_{916})) + 1)) \wedge ((\exists v_{919}((v_{912} + S(v_{919})) = S((S(v_{906} \cdot 60) \cdot v_{916}))) \wedge \exists v_{920}(v_{915} = ((v_{920} \cdot S((S(v_{906} \cdot 60) \cdot v_{916}))) + v_{912}))) \wedge \forall v_{921}(\exists v_{922}((v_{921} + S(v_{922})) = (v_{906} \cdot 60) \cdot v_{916}))) \wedge \exists v_{923} \exists v_{924}((\exists v_{925}((v_{923} + S(v_{925})) = S((S(v_{921}) \cdot v_{916}))) \wedge \exists v_{926}(v_{915} = ((v_{926} \cdot S((S(v_{921}) \cdot v_{916}))) + v_{923}))) \wedge ((\exists v_{927}((v_{924} + S(v_{927})) = S((S(v_{921}) \cdot v_{916}))) \wedge \exists v_{928}(v_{915} = ((v_{928} \cdot S((S(v_{921}) \cdot v_{916}))) + v_{924}))) \wedge (v_{924} = (v_{923} + v_{923})))))) \wedge ((v_{611} = ((v_{913} \cdot (v_{912} \cdot 1152921504606846976) + (v_{910} \cdot v_{912}) + v_{914}))) \wedge \exists v_{929}((v_{914} + S(v_{929})) = v_{912}))) \wedge (\exists v_{930} \exists v_{931} \exists v_{932}(\exists v_{933} \exists v_{934}((\exists v_{935}((1 + S(v_{935})) = S(v_{934})) \wedge \exists v_{936}(v_{933} = ((v_{936} \cdot S(v_{934})) + 1)) \wedge ((\exists v_{937}((v_{930} + S(v_{937})) = S((S(v_{906} \cdot 60) \cdot v_{934}))) \wedge \exists v_{938}(v_{933} = ((v_{938} \cdot S((S(v_{906} \cdot 60) \cdot v_{934}))) + v_{930}))) \wedge \forall v_{939}(\exists v_{940}((v_{939} + S(v_{940})) = (v_{906} \cdot 60) \cdot v_{934}))) \wedge \exists v_{941} \exists v_{942}((\exists v_{943}((v_{941} + S(v_{943})) = S((S(v_{939}) \cdot v_{934}))) \wedge \exists v_{944}(v_{933} = ((v_{944} \cdot S((S(v_{939}) \cdot v_{934}))) + v_{941}))) \wedge ((\exists v_{945}((v_{942} + S(v_{945})) = S((S(v_{939}) \cdot v_{934}))) \wedge \exists v_{946}(v_{933} = ((v_{946} \cdot S((S(v_{939}) \cdot v_{934}))) + v_{942}))) \wedge (v_{942} = (v_{941} + v_{941})))))) \wedge ((v_{616} = ((v_{931} \cdot (v_{930} \cdot 1152921504606846976) + (v_{911} \cdot v_{930}) + v_{932}))) \wedge \exists v_{947}((v_{932} + S(v_{947})) = v_{930}))) \wedge \exists v_{948} \exists v_{949} \exists v_{950} \exists v_{951}((v_{910} = ((v_{948} \cdot v_{841}) + ((v_{949} \cdot v_{840}) + v_{951}))) \wedge ((v_{911} = ((v_{948} \cdot v_{841}) + ((v_{950} \cdot v_{840}) + v_{951}))) \wedge (((v_{948} = 2) \wedge (v_{949} = v_{610})) \rightarrow (v_{950} = v_{617})) \wedge (\neg((v_{948} = 2) \wedge (v_{949} = v_{610})) \rightarrow (v_{950} = v_{949})))))) \wedge \forall v_{952} \exists v_{953}(\exists v_{954}((v_{952} + S(v_{954})) = v_{194}) \wedge (\exists v_{955}((v_{953} + S(v_{955})) = v_{194}) \wedge \exists v_{956} \exists v_{957}((\exists v_{958}((v_{956} + S(v_{958})) = S((S(v_{952}) \cdot v_{190}))) \wedge \exists v_{959}(v_{189} = ((v_{959} \cdot S((S(v_{952}) \cdot v_{190}))) + v_{956}))) \wedge ((\exists v_{960}((v_{957} + S(v_{960})) = S((S(v_{953}) \cdot v_{190}))) \wedge \exists v_{961}(v_{189} = ((v_{961} \cdot S((S(v_{953}) \cdot v_{190}))) + v_{957}))) \wedge \exists v_{962} \exists v_{963} \exists v_{964} \exists v_{965} \exists v_{966} \exists v_{967} \exists v_{968}(\exists v_{969} \exists v_{970} \exists v_{971}(\exists v_{972} \exists v_{973}((\exists v_{974}((1 + S(v_{974})) = S(v_{973})) \wedge \exists v_{975}(v_{972} = ((v_{975} \cdot S(v_{973})) + 1)) \wedge ((\exists v_{976}((v_{969} + S(v_{976})) = S(v_{973})) \wedge \exists v_{977}(v_{972} = ((v_{977} \cdot S(v_{973})) + v_{969}))) \wedge \forall v_{978}(\exists v_{979}((v_{978} + S(v_{979})) = 0) \rightarrow \exists v_{980} \exists v_{981}(\exists v_{982}((v_{980} + S(v_{982})) = S((S(v_{978}) \cdot v_{973}))) \wedge \exists v_{983}(v_{972} = ((v_{983} \cdot S((S(v_{978}) \cdot v_{973}))) + v_{980}))) \wedge ((\exists v_{984}((v_{981} + S(v_{984})) =$

